

# Confidence-Aware Retrieval-Augmented Generation with Abstention for Cyber Threat Intelligence

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**Abstract**—Retrieval-Augmented Generation (RAG) grounds language models in external knowledge, which makes it a natural tool for Cyber Threat Intelligence (CTI). Standard RAG, however, answers every question, even when the retrieved evidence does not support an answer, and can mislead an analyst with invented vulnerability details. We study a confidence-aware RAG pipeline for CVE search that declines to answer when a retrieval-stage confidence score falls below a threshold. The score combines the similarity of the best-matching record with its margin over the second record, and needs no access to the language model’s internals. On 200 automatically generated keyword queries over the 2023 National Vulnerability Database (30,932 CVE records), the gate abstains on 16% of queries and raises precision on answered queries to 75.6%, against 70.0% for BM25-only RAG and 73.7% for the same hybrid pipeline without the gate. The rejected queries are genuinely hard: the ungated system answers only 31.3% of them correctly. Raising the threshold trades coverage for precision, up to 85.0% at 20% coverage. We also show that the score ranks answers by reliability (AUROC 0.70) but is not a calibrated probability, and we explain why Expected Calibration Error favours the ungated baseline.

**Index Terms**—Retrieval-Augmented Generation, Cyber Threat Intelligence, confidence estimation, abstention, selective prediction, CVE, NVD.

## I. INTRODUCTION

Cyber Threat Intelligence (CTI) analysts spend much of their day searching large vulnerability databases, such as the National Vulnerability Database (NVD) [19], for precise information. In this setting a wrong answer is expensive. A single incorrect CVE reference can lead an analyst to patch the wrong software, miss a real exposure, or misdirect an incident response. A

system that answers confidently but incorrectly is therefore more harmful than one that admits it does not know.

Retrieval-Augmented Generation (RAG) [1] is a natural fit for this task. It retrieves relevant documents and lets a language model write an answer grounded in them. Standard RAG pipelines never check whether the retrieved evidence actually supports the question. When retrieval fails, the model still produces a fluent but wrong answer. This is especially harmful in CVE-based CTI, where severity scores, affected versions, and weakness categories must be exact.

Most ways of estimating confidence do not fit this setting. Some read the model’s token probabilities [16], which small, locally hosted models often do not expose. Others measure uncertainty only in the generated text [13] and ignore the retrieval stage, where the uncertainty in CVE search begins. Methods that act at retrieval time, such as REPLUG [4], reweight documents but give no signal for deciding when the evidence is too weak to answer at all.

**Contributions:** This paper makes four contributions.

- A retrieval-only abstention gate for CTI question answering. It decides before generation, so rejected queries cost no language-model call and need no model internals.
- An analysis of which retrieval signal carries the information. In hybrid retrieval the most useful signal is agreement between the lexical and dense retrievers, while the margin term adds little.
- A precision-coverage study over the full query set, with paired significance tests, bootstrap confidence intervals, and

a quantitative analysis of over-abstention and over-confident errors.

- A careful treatment of calibration for generative retrieval. We define how Expected Calibration Error (ECE) is computed with abstention and show why it can reward an uninformative baseline.

The rest of this paper is organized as follows. Section II reviews related work. Section III presents the framework. Section IV describes the experimental setup. Section V reports the results. Section VI discusses failure modes and generalizability, and Section VII concludes the paper.

## II. RELATED WORK

Table I groups the most relevant prior work and shows where our system fits.

*RAG and adaptive retrieval:* The original RAG framework [1] always retrieves a fixed number of documents. Self-RAG [2] learns when to retrieve and critiques its own output, but it needs a specially trained generator. CRAG [3] scores retrieval quality with a trained evaluator and, when it is low, corrects the context with web search instead of abstaining. REPLUG [4] reweights documents for a frozen model but never declines to answer.

*Retrieval confidence and selective prediction:* Using retrieval scores to predict whether a search will succeed is an old idea in information retrieval, known as query performance prediction [10]. Abstaining when confidence is low is studied as selective prediction [11] and, for question answering, as selective QA [12]; both evaluate with risk-coverage curves, which we adopt. Our score is a simple member of this family, so we do not claim a new uncertainty estimator. Our contribution is to use such a score as a pre-generation gate for CTI with a local model, to identify which signal drives it in hybrid retrieval, and to evaluate the resulting trade-off reproducibly. Dense retrievers [5] and BM25 [6] are complementary, and reciprocal rank fusion [7] is a standard way to combine them [8]. Semantic entropy [13] and self-evaluation [14] estimate uncertainty from the generator instead, and CySecBERT [17] adapts encoders to security text. CTI surveys flag relevance filtering as an open problem [19].

## III. PROPOSED FRAMEWORK

Fig. 1 gives an overview. A query passes through hybrid retrieval, a confidence score is computed from the retrieval scores, and a threshold decides whether the system answers or abstains.

### A. Dataset

Our knowledge base is the 2023 NVD JSON feed (version 2.0). We drop records whose description is shorter than 50 characters, which leaves 30,932 entries. Each record carries a CVE identifier, a CVSS severity, a CWE category, and a free-text description. Before indexing, we prepend a short tag with the CVE identifier, severity, and CWE code to each description, which helps BM25 match exact identifiers.

### B. Problem Formulation

Given a query and the corpus, retrieval returns the top- $K$  records, a confidence step scores the retrieval with a value  $C \in [0, 1]$ , and a generator writes an answer from the records. A threshold  $\theta$  controls the system:

$$\text{decision} = \begin{cases} \text{answer}, & C \geq \theta, \\ \text{abstain}, & C < \theta. \end{cases} \quad (1)$$

When the system abstains, it returns an “Insufficient context” message and does not call the language model.

### C. Hybrid Retrieval

*Sparse retrieval:* A BM25 index [6] over the tagged descriptions recalls exact identifiers, CWE codes, and product names. Tokenisation strips punctuation but keeps hyphens.

*Dense retrieval:* We encode every record with all-MiniLM-L6-v2 [9] and search the normalised 384-dimensional vectors exactly with FAISS (inner product, equal to cosine similarity).

*Reciprocal rank fusion:* Each retriever returns  $2K$  candidates, which we merge with reciprocal rank fusion [7]:

$$\text{RRF}(d) = \sum_{r \in \{r_{\text{BM25}}, r_{\text{dense}}\}} \frac{1}{k + \text{rank}_r(d)}, \quad k = 60. \quad (2)$$

The constant  $k = 60$  is the value recommended in [7]. The top  $K = 5$  fused records go to the confidence step.

### D. Confidence Scoring

Let  $s_1$  and  $s_2$  be the dense cosine similarities of the first and second fused records. The confidence score is

$$C = \alpha s_1 + (1 - \alpha) m, \quad m = \frac{s_1 - s_2}{s_1}, \quad \alpha = 0.6. \quad (3)$$

The first term reflects raw match quality. The margin  $m$  reflects how clearly the top record stands out; a near tie suggests an ambiguous query. A record that only BM25 retrieved has no dense similarity and is assigned a similarity of zero. When such a record is ranked first,  $C = 0$ . This case means the two retrievers disagree about the best record, and it turns out to be the strongest signal in our data (Section V).

*Parameter choice:* We fixed  $\alpha = 0.6$  and  $\theta = 0.20$  during development on the same query distribution; there was no separate validation split. We therefore report sensitivity to both parameters in Section V rather than presenting them as tuned optima.

### E. Answer Generation

When the gate allows an answer, the five top records are passed to Llama-3.2 (3B), run locally through Ollama [20], with greedy decoding and a 512-token limit. The prompt tells the model to use only the supplied records, to say so when they are insufficient, and to cover the vulnerability, affected component, severity, and impact. Algorithm 1 summarises the pipeline.

*Worked example:* For the query “Apache HTTP/2 rapid reset denial of service”, both retrievers agree on CVE-2023-44487, and  $C = 0.36$ , so the system answers. The vague query “recent

TABLE I  
REPRESENTATIVE RELATED WORK AND POSITIONING OF OUR APPROACH.

| Ref        | Study                  | Yr         | Approach / contribution                                                 | Limitation we address                 |
|------------|------------------------|------------|-------------------------------------------------------------------------|---------------------------------------|
| [1]        | Lewis et al.           | 2020       | DPR + BART RAG; retrieval and generation integration                    | Fixed top- $K$ ; no confidence signal |
| [2]        | Asai et al.            | 2024       | Self-RAG: adaptive retrieval with reflection tokens                     | Requires a fine-tuned generator       |
| [3]        | Yan et al.             | 2024       | CRAG: retrieval evaluator triggers correction and web search            | Corrects rather than abstains         |
| [4]        | Shi et al.             | 2024       | REPLUG: black-box reweighting by similarity                             | No abstention decision                |
| [10]       | Cronen-Townsend et al. | 2002       | Query performance prediction from retrieval scores                      | Not used to gate generation           |
| [11], [12] | Geifman; Kamath et al. | 2017; 2020 | Selective prediction and selective QA with risk-coverage analysis       | Not studied for CTI retrieval         |
| [15]       | Guo et al.             | 2017       | Expected Calibration Error (ECE)                                        | Defined for classification only       |
| [18]       | Tihanyi et al.         | 2024       | CyberMetric: cybersecurity LLM benchmark                                | Closed-book; no retrieval gate        |
| This work  |                        | 2026       | Retrieval-only abstention gate for NVD search; precision-coverage study | Keyword queries; single corpus        |

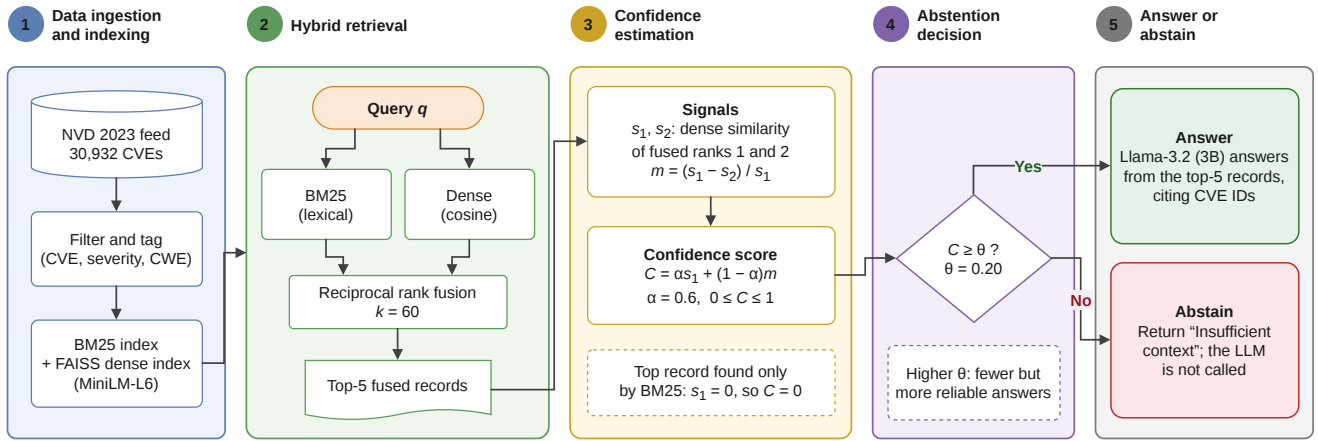


Fig. 1. Overview of the Confidence-Aware RAG pipeline. Hybrid BM25, dense, and reciprocal-rank-fusion retrieval feed a retrieval-stage confidence score  $C$ . A threshold  $\theta$  then decides between answering and abstaining.

#### Algorithm 1 Confidence-Aware RAG with Abstention

**Require:** query  $q$ , corpus  $\mathcal{C}$ , threshold  $\theta$ , top- $K$ , weight  $\alpha$   
**Ensure:** answer  $a$  or an abstention signal

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1:  $R_{\text{BM25}} \leftarrow \text{BM25}(q, \mathcal{C}, 2K)$ 
2:  $R_{\text{dense}} \leftarrow \text{DenseSearch}(q, \mathcal{C}, 2K)$ 
3:  $R \leftarrow \text{RRF}(R_{\text{BM25}}, R_{\text{dense}}, k=60)[1:K]$ 
4: get dense similarities  $s_1, s_2$  of the top two records in  $R$ 
5:  $m \leftarrow (s_1 - s_2) / s_1$ ;  $C \leftarrow \alpha s_1 + (1 - \alpha)m$ 
6: if  $C \geq \theta$  then
7:    $a \leftarrow \text{LLM}(q, R, \text{temp}=0, \text{max\_tok}=512)$ 
8:   return  $a$  with source citation
9: else
10:  return "Insufficient context"
11: end if

```

Linux bug" gives two records with identical similarity (0.54), so the margin is zero and  $C = 0.33$ ; the system answers at  $\theta = 0.20$  but would abstain at  $\theta \geq 0.35$ . For "Incorrect X-Forwarded-For pretix Incorrect parsing", BM25 ranks the correct record first but the dense retriever does not return it, so  $C = 0$  and the system abstains even though the answer was available. Section VI analyses such cases.

#### IV. EXPERIMENTAL SETUP

**Queries and gold answers:** We sample 200 CVE records uniformly at random (seed 42). For each record, a query of up to five keywords is extracted automatically from its description with one of three templates: technical terms (51.5% of queries), severity plus vulnerability type (28.5%), and product names (20.0%). CVE identifiers are removed from the query. Examples are "Linux kernel rose\_ioctl" and "MEDIUM Scripting Cross-Site Scripting 3.1.6 Auth". This is a known-item setting: each query has one target record, whose NVD description serves as the gold answer. The queries are not written by analysts, and because they reuse words from the target record they favour lexical retrieval. The results therefore measure confidence-based selective answering in a controlled, reproducible CTI retrieval setting, not performance on unrestricted analyst queries (Section VI).

**Systems compared:** Naive RAG uses BM25 only and never abstains. Hybrid RAG adds dense search and fusion and never abstains. Confidence-Aware RAG is Hybrid RAG with the gate at  $\theta = 0.20$ . All three use the same generator and prompt. The

TABLE II

RESULTS ON 200 QUERIES AT  $\theta = 0.20$  (C IS THE PROPOSED SYSTEM).  
R-L: ROUGE-L OVER ALL QUERIES AND OVER ANSWERED QUERIES.  
PREC: PRECISION@ANSWERED. COV: COVERAGE.

| System        | Retr. | Answer quality |          | Reliability |      | Confidence |       |
|---------------|-------|----------------|----------|-------------|------|------------|-------|
|               | R@5   | R-L            | R-L(ans) | Prec        | Cov  | ECE        | AUROC |
| A. Naive RAG  | 0.810 | 0.330          | 0.330    | 0.700       | 100% | 0.038      | 0.690 |
| B. Hybrid RAG | 0.795 | 0.298          | 0.321    | 0.737       | 93%  | 0.309      | 0.700 |
| C. Conf-Aware | 0.795 | 0.280          | 0.334    | 0.756       | 84%  | 0.259      | 0.700 |

language model may still refuse on its own; Hybrid RAG did so for 14 queries.

*Evaluation metrics:* We group the metrics by what they measure. Retrieval is measured by Recall@5, the share of queries whose target record is in the top five. Answer quality is ROUGE-L F1 against the gold description, over all queries (abstentions score zero) and over answered queries. An answer counts as correct when its ROUGE-L is at least 0.20, and reliability is Precision@Answered, the share of answered queries that are correct. Coverage is the share of queries answered.

*Confidence quality:* We report two measures, both using the correctness label above. Discrimination is the AUROC of  $C$  for separating correct from incorrect answers. Calibration is ECE with ten equal-width bins on  $[0, 1]$  [15]:  $ECE = \sum_b \frac{|B_b|}{n} |\text{acc}(B_b) - \text{conf}(B_b)|$ , where  $\text{acc}$  is the share of correct answers in bin  $B_b$  and  $\text{conf}$  its mean confidence. In Table II ECE is taken over all 200 queries, and an abstained query keeps its confidence and counts as incorrect. Naive RAG has no dense retriever, so the same formula is applied with BM25 scores divided by the top BM25 score.

*Statistics and implementation:* ROUGE-L differences are tested with paired  $t$ -tests on per-query scores; with  $n = 200$  the mean difference is approximately normal, and we confirm each result with a Wilcoxon signed-rank test. Precision compares different subsets of queries, so we use a paired bootstrap with 10,000 resamples. All experiments run on an Apple MacBook Air (M-series, 16 GB RAM) with fixed seeds. The gate does not change the answer to an accepted query, so each threshold in Section V is evaluated by gating the saved Hybrid RAG answers.

## V. RESULTS AND ANALYSIS

### A. System Comparison

Table II groups the results by retrieval, answer quality, reliability, and confidence quality.

*Retrieval and answer quality:* BM25 alone gives the highest Recall@5 (0.810), as expected for keyword queries built from the target record. The gate does not change retrieval. Over all queries Confidence-Aware RAG has lower ROUGE-L than Naive RAG, because each abstention scores zero (mean difference  $-0.050$ ,  $t(199) = -5.12$ ,  $p < 0.001$ ,  $d_z = -0.36$ , 95% CI  $[-0.068, -0.031]$ ). On the 168 queries it answers, its ROUGE-L does not differ significantly from Naive RAG’s

TABLE III

ABLATION OF THE CONFIDENCE SCORE (200 QUERIES).  
PRECISION@ANSWERED AT MATCHED COVERAGE.

| Score variant                 | AUROC | Prec @ 84% cov.   | Prec @ 50% cov. |
|-------------------------------|-------|-------------------|-----------------|
| Full, $\alpha = 0.6$ (ours)   | 0.700 | 0.756             | 0.780           |
| Similarity only, $\alpha = 1$ | 0.699 | 0.756             | 0.810           |
| Margin only, $\alpha = 0$     | 0.637 | 0.708             | 0.770           |
| No gate (Hybrid RAG)          | n/a   | 0.737 at 93% cov. |                 |

answers to the same queries ( $-0.014$ ,  $t(167) = -1.75$ ,  $p = 0.08$ ; Wilcoxon  $p = 0.58$ ).

*Reliability and coverage:* Confidence-Aware RAG answers 84% of queries with 75.6% precision, against 70.0% for Naive RAG and 73.7% for Hybrid RAG. The 5.6-point gain over Naive RAG is suggestive but not significant at this sample size (paired bootstrap 95% CI  $[-0.8, 12.2]$ ). The clearer evidence is what the gate removes. On the 32 rejected queries, the ungated Hybrid RAG is correct only 31.3% of the time, against 75.6% on the accepted ones (Fisher’s exact test, odds ratio 6.8,  $p < 10^{-5}$ ). All 14 queries on which the generator refused by itself also have  $C = 0$ , so the gate catches them before any model call.

*Calibration:* Naive RAG has by far the lowest ECE (0.038), but this does not mean it is better calibrated in a useful sense. Its score lies between 0.60 and 0.94 for every query (mean 0.67), close to its overall accuracy (0.70), so a nearly constant score matches the average accuracy. ECE rewards this, yet the score barely separates right from wrong answers (AUROC 0.69). Our score has a mean of 0.38 against an accuracy near 0.70, so it is not a calibrated probability, which explains its high ECE. Most of the drop from 0.309 to 0.259 between Hybrid and Confidence-Aware RAG comes from the abstained queries (score 0, counted incorrect); on answered queries alone ECE is 0.333 and 0.309. We therefore treat  $C$  as a ranking signal for selective answering, and AUROC and precision-coverage, rather than ECE, as the relevant measures.

### B. Ablation of the Confidence Score

Table III compares the full score with each term alone, ranking all 200 queries and answering the top 84% or 50%.

The top-1 similarity carries almost all of the signal; the margin alone is clearly weaker, and adding it to the similarity does not help. At 84% coverage the full and similarity-only scores make identical decisions, because both reject exactly the 32 queries with  $C = 0$ . Varying  $\alpha$  from 0.5 to 1.0 changes AUROC only between 0.69 and 0.72, so the choice of  $\alpha$  is not critical.

### C. Threshold Sensitivity

Table IV and Fig. 2 vary the threshold on all 200 queries. Every threshold up to 0.22 gives the same decisions as 0.20, because the smallest non-zero score is 0.23. Above that, precision rises steadily as coverage falls, from 75.6% at 84% coverage to 85.0% at 20% coverage. No single threshold is best: 0.20 suits a first-pass filter, while higher values suit settings

TABLE IV  
THRESHOLD SENSITIVITY ON 200 QUERIES.

| $\theta$ | Coverage | Answered | Prec@Answered | R-L(ans) |
|----------|----------|----------|---------------|----------|
| none     | 93.0%    | 186      | 0.737         | 0.321    |
| 0.20     | 84.0%    | 168      | 0.756         | 0.334    |
| 0.30     | 75.0%    | 150      | 0.780         | 0.340    |
| 0.40     | 42.5%    | 85       | 0.800         | 0.363    |
| 0.50     | 20.0%    | 40       | 0.850         | 0.385    |

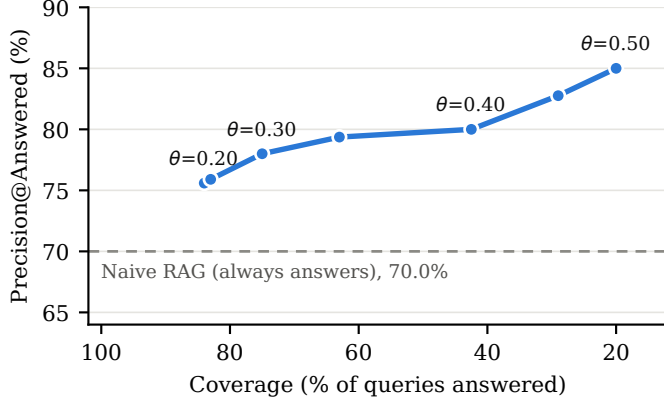


Fig. 2. Precision on answered queries against coverage as the threshold  $\theta$  rises (200 queries). The dashed line is Naive RAG, which always answers.

where an analyst prefers fewer but more reliable answers. With 200 queries and no held-out split, these values should be re-checked on the target query distribution.

## VI. DISCUSSION

**Over-abstention:** Of the 32 rejected queries, 10 would have been answered correctly, and for 16 the target record was in the top five. In these cases BM25 finds the record but the dense retriever does not, as in the prefix example of Section III. Rejections are spread evenly over the three query types (11, 11, and 10). Many rejected queries are strings of version numbers, such as “14.5 14.6 14.7 HTTP CVSS”, where abstaining is reasonable.

**Over-confident errors:** Of the 41 incorrect answers, the target record was missing from the top five in only 9. Among the 40 answers with  $C \geq 0.5$ , 6 are counted incorrect, and in all 6 the target was retrieved. For “Linux kernel rose\_ioctl” ( $C = 0.82$ ) the answer correctly names CVE-2023-51782 and describes the use-after-free, but its ROUGE-L is 0.07 because it is formatted as a list. Many such errors thus reflect the metric rather than the system: a lexical match against the full NVD description penalises correct answers with different wording. Product queries are the hardest (46.7% precision against 79.3% and 87.0% for the other types).

**Generalizability:** Three features of the setup limit how far the numbers transfer. The keyword queries share words with their target, which favours BM25 and inflates recall compared with questions written by analysts. NVD records are short and uniform, whereas real CTI work mixes advisories, threat

reports, and exploit data of varied length and quality, where retrieval scores behave differently. The gold answers are lexical, so correctness is measured imperfectly. The qualitative finding, that a retrieval-only score can separate reliable from unreliable answers before generation, is likely to hold more broadly, but the thresholds and the size of the gain must be re-measured on each deployment’s data. A direct comparison with Self-RAG and CRAG on the same corpus, model, and prompts is also left for future work.

## VII. CONCLUSION

We studied a confidence-aware RAG pipeline for CVE search that abstains when a retrieval-stage score is low. On 200 queries over 30,932 NVD records, the gate removes queries that the ungated system mostly gets wrong, raises precision on answered queries from 73.7% to 75.6% at 84% coverage, and reaches 85.0% at 20% coverage. The score works as a ranking signal, not a calibrated probability, and its strongest component is agreement between the lexical and dense retrievers. Future work will use natural-language analyst queries with a held-out split, factual rather than lexical answer evaluation, heterogeneous CTI sources, and direct comparison with generator-side methods. Code, queries, and per-query results are available at <https://github.com/latifurrafi/CTI-RAG>.

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